



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2009

Yearly Examination

YEAR 11

Mathematics (2 unit) & Extension Continuers

Instructions:

- Each Question is to be returned in a separate booklet.
- **Question 1 & 2 are to be collected after 60 minutes** at which time the 2 unit Mathematics students will be dismissed.
- **Question 3 & 4 are to be collected after 105 minutes.**
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise stated.
- Full marks may not be awarded for careless or badly arranged work.

Examiner: *A Fuller*

(Use a SEPARATE writing booklet)

Question 1 (28 marks)

(a) Find the value of $\log_3 9$.

1

(b) Solve the following for x :

6

(i) $\log_6 x = 3$

(ii) $\log_x 3 = -1$

(iii) $2^{2x+1} = \frac{1}{16}$

(iv) $x^2 + 3x - 18 = 0$

(c) Differentiate the following with respect to x :

7

(i) $2x + 5$

(ii) $\frac{1}{2x + 5}$

(iii) $\frac{2x + 5}{x}$

(iv) $\frac{x}{2x + 5}$

- (d) Sketch the following on separate axes showing any intercepts with the co-ordinate axes and any asymptotes:

4

(i) $y = \frac{1}{x} + 1$

(ii) $y = \log_2(x + 1)$

- (e) Write $2x^2 - 7x - 4$ in the form $a(x + 2)^2 + b(x + 2) + c$.

3

- (f) Consider the arithmetic series: $-1 + 3 + 7 + 11 + 15 + \dots$

4

- (i) Which term of the series is 391?

- (ii) Hence, find the sum up to the term which is 391.

- (g) If $f(x) = x^3 - 3x^2 - 6x$. Evaluate:

3

(i) $f(-2)$

(ii) $f'(2)$

(Use a SEPARATE writing booklet)

Question 2 (28 marks)

(a) Consider the geometric series: $27 + 18 + 12 + 8 + \dots$

2

(i) Explain why the series has a limiting sum.

(ii) Find the limiting sum of the series.

(b) Find the co-ordinates of the focus and the equation of the directrix of the parabola $y = \frac{x^2}{4} - 1$.

3

(c) State whether the following functions are odd, even, or neither:

3

(i) $f(x) = x^6 + 10$

(ii) $f(x) = \frac{x^2}{2 - x}$

(iii) $f(x) = \log_{10} 2^x$

(d) Find using first principles the derivative $f'(x)$ given that $f(x) = x^3$.

2

(e) Let $\log_5 3 = a$ and $\log_5 2 = b$.

4

(i) Find the following in terms of a and b :

(α) $\log_5 6$

(β) $\log_5 \left(\frac{1}{4}\right)$

(ii) Evaluate 5^{2a} .

(f) Find the value(s) of k for which $x^2 - kx + 4 = 0$ has:

5

(i) one root equal to -1

(ii) real roots

(iii) one root double the other.

(g) Find the domain and the range of the following:

4

(i) $y = \sqrt{1 - x}$

(ii) $y = \sqrt{1 - x^2}$

- (h) Caleb plans to deposit an amount of money into an account which will pay him 1% interest each month on the balance of his account at the time.

Immediately after each interest payment is made, Caleb plans to withdraw \$1000.

Let his deposit be $\$D$.

- (i) Show that when he has made his second withdrawal, the balance of his account will be $\$[D(1.01)^2 - 1000(1 + 1.01)]$.

- (ii) Caleb wants his deposit to be sufficient to be able to make withdrawals for 10 years. Find, to the nearest \$100, what his deposit must be.

(Use a SEPARATE writing booklet)

Question 3 (18 marks)

(a) Find the point dividing the interval from $(-3, 4)$ to $(5, -2)$ in the ratio $1 : 3$. 2

(b) Find the acute angle between the lines $y = 3x - 2$ and $x + 2y - 3 = 0$. 3

Give the answer to the nearest degree.

(c) (i) Show that $(x + 2)$ is a factor of $6x^3 + 7x^2 - 9x + 2$. 3

(ii) Hence, or otherwise find all of the factors of $6x^3 + 7x^2 - 9x + 2$.

(d) Find the general solution for $\tan \theta + 1 = 0$ (in radians). 3

(e) (i) Write $\cos \theta - \sqrt{3} \sin \theta$ in the form $R \cos(\theta + \alpha)$. 4

(ii) Hence, or otherwise, solve $\cos \theta - \sqrt{3} \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$.

(f) Given that $\sin \theta = \frac{1}{\sqrt{3}}$ and $\frac{\pi}{2} < \theta < \pi$. Find the exact value of the following: 3

(i) $\tan \theta$

(ii) $\cos 2\theta$

(Use a SEPARATE writing booklet)

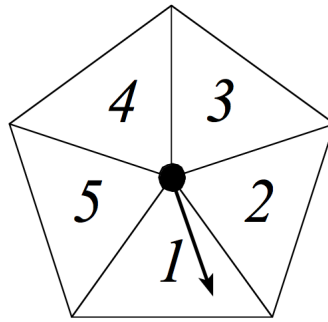
Question 4 (17 marks)

(a) Solve $\frac{x+4}{x-2} \geq 3$.

3

(b)

6



(i) The arrow on the regular pentagon is spun twice and the sum of the two numbers is recorded. Find the probability of getting:

(α) an odd result

(β) a result of at least 7

(ii) How many times must the arrow on the regular pentagon be spun to be 99.9% sure of getting at least one 5?

(c) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The tangents at P and Q intersect at the point T . Let $S(0, a)$ be the focus of the parabola.

(i) Show that the equation of the tangent to the parabola at P is given by

$$y = px - ap^2.$$

(ii) Find the co-ordinates of T .

(iii) Show that $SP = a(p^2 + 1)$.

(iv) Suppose P and Q move on the parabola so that $SP + SQ = 4a$.

Show that the locus of the point T is a parabola.

End of paper

$$(a) \log_3 9 = 2 \quad [1]$$

$$(b) (i) \log_6 x = 3$$

$$x = 6^3$$

$$x = 216 \quad [1]$$

$$(ii) \log_x 3 = -1$$

$$3 = x^{-1}$$

$$x = \frac{1}{3} \quad [1]$$

$$(iii) 2^{2x+1} = \frac{1}{16}$$

$$2^{2x+1} = 2^{-4}$$

$$2x+1 = -4$$

$$x = -2\frac{1}{2} \quad [2]$$

$$(iv) x^2 + 3x - 18 = 0$$

$$(x+6)(x-3) = 0$$

$$x = -6 \text{ or } 3 \quad [2]$$

$$(c) (i) \frac{d}{dx}(2x+5) = 2 \quad [1]$$

$$(ii) \frac{d}{dx}\left(\frac{1}{2x+5}\right) = \frac{d}{dx}((2x+5)^{-1})$$

$$= -1 \times (2x+5)^{-2} \times 2$$

$$= \frac{-2}{(2x+5)^2} \quad [2]$$

$$(iii) \frac{d}{dx}\left(\frac{2x+5}{x}\right) = \frac{d}{dx}\left(2 + \frac{5}{x}\right)$$

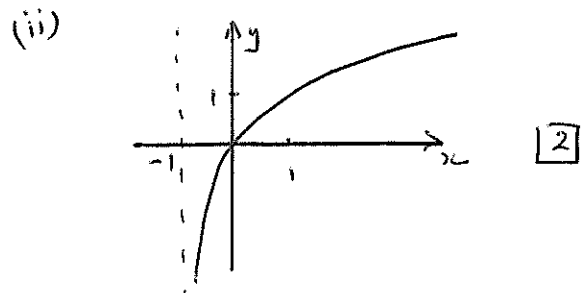
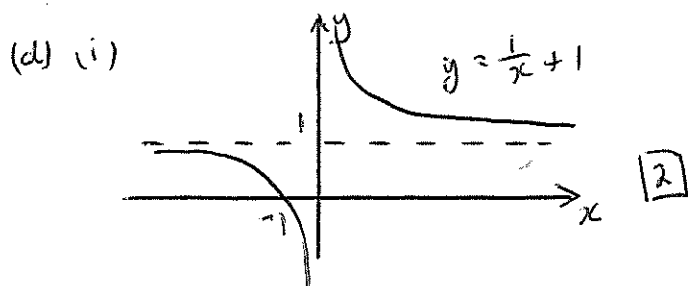
$$= 5x^{-1} - 1x^{-2}$$

$$= \frac{-5}{x^2} \quad [2]$$

$$(iv) \frac{d}{dx}\left(\frac{x}{2x+5}\right) = \frac{(2x+5) \cdot 1 - x \cdot 2}{(2x+5)^2}$$

$$= \frac{2x+5-2x}{(2x+5)^2}$$

$$= \frac{5}{(2x+5)^2} \quad [2]$$



(e) $2x^2 - 7x - 4 = a(x+2)^2 + b(x+2) + c$

Equating leading coefficients $a = 2$

Let $x = -2$: $8 + 14 - 4 = c$

$$\therefore c = 18$$

Let $x = 0$: $-4 = 2 \times 4 + b \times 2 + 18$

$$-4 = 2b + 26$$

$$2b = -30$$

$$b = -15 \quad [3]$$

$$2x^2 - 7x - 4 = 2(x+2)^2 - 15(x+2) + 18$$

(f) $a = -1$

$$d = 4$$

(i) $T_n = -1 + (n-1) \cdot 4 = 391$

$$4(n-1) = 392$$

$$n-1 = 98$$

$$n = 99 \quad [2]$$

(ii) $S_{99} = \frac{99}{2}(-1 + 391)$

$$= \frac{99}{2} \times 390$$

$$= 19305 \quad [2]$$

(g) (i) $f(-2) = -8 - 12 + 12$

$$= -8 \quad [1]$$

(ii) $f'(x) = 3x^2 - 6x - 6$

$$f'(2) = 12 - 12 - 6$$

$$= -6 \quad [2]$$

28

Question 2.

$$f(-x) = \frac{(-x)^2}{2 - (-x)}$$
$$= \frac{x^2}{2 + x}$$

a) If $|r| < 1$, the sum of
① the series is limited to
some finite number ①

$\therefore f(x)$ is neither ①

ii) $a = 27$ $r = \frac{2}{3}$

iii) $f(x) = \log_{10} 2^x = x \log_{10} 2$

$$S_{\infty} = \frac{27}{1 - \frac{2}{3}}$$

$$= 81$$

①

$$f(-x) = \log_{10} 2^{-x}$$

$$= -x \log_{10} 2$$

$$= -f(x)$$

b) $y = \frac{x^2}{4} - 1$

$\therefore f(x)$ is odd. ①

$$4y = x^2 - 4$$

$$x^2 = 4y + 4$$

$$x^2 = 4(y + 1)$$

d) $f'(x) = \frac{f(x+h) - f(x)}{h}$

$$= \frac{(x+h)^3 - x^3}{h}$$

$\therefore h = 0, k = -1, a = 1$ ①

focus: $(0, 0)$

①

$$(x+h)^3 = (x^2 + 2xh + h^2)(x+h)$$
$$= x^3 + x^2h + 2x^2h + 2xh^2$$
$$+ h^2x + h^3$$

$$= x^3 + 3x^2h + 3xh^2 + h^3$$

directrix: $y = -1 - 1$

$$y = -2$$

①

$$= \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$$

$$= \frac{3x^2h + 3xh^2 + h^3}{h}$$

$$= \cancel{h} (3x^2 + 3xh + h^2)$$
$$\cancel{h}$$

$\therefore f(x)$ is even ①

$$\lim_{h \rightarrow 0} = 3x^2$$

②

ii) $f(x) = \frac{x^2}{2-x}$

$$h \rightarrow 0$$

$$\therefore f'(x) = 3x^2$$

$$e) \log_5 3 = a$$

$$\log_5 2 = b$$

$$i) \alpha) \log_5 6 = \log_5 3 + \log_5 2$$

$$= a + b \quad (1)$$

$$\beta) \log_5 (1/4) = \log_5 1 - \log_5 4$$

$$= 0 - \log_5 2^2$$

$$= 0 - 2\log_5 2$$

$$= -2b \quad (1)$$

$$ii) 5^{2a} = (5^a)^2$$

$$= 3^2$$

$$= 9 \quad (2)$$

$$f) x^2 - kx + 4 = 0$$

$$i) (-1)^2 + k + 4 = 0$$

$$k = -5 \quad (1)$$

$$ii) \Delta > 0$$

$$\therefore b^2 - 4ac > 0$$

$$b^2 - 4ac > 0$$

$$(-k)^2 - 4 \times 1 \times 4 > 0$$

$$k^2 - 16 > 0$$

$$k^2 > 16$$

$$k > 4, k < -4 \quad (2)$$

$$iii) \alpha + \beta = -b/a$$

$$\alpha + 2\alpha = k$$

$$3\alpha = k$$

$$\alpha \times 2\alpha = 4$$

$$2\alpha^2 = 4$$

$$\alpha^2 = 2$$

$$\alpha = \pm \sqrt{2}$$

$$\therefore k = 3\sqrt{2}, -3\sqrt{2} \quad (2)$$

$$g) i) D: x \leq 1 \quad (1)$$

$$R: y \geq 0 \quad (1)$$

$$ii) D: -1 \leq x \leq 1 \quad (1)$$

$$R: 0 \leq y \leq 1 \quad (1)$$

h) i) After 1 month

$$A_1 = (\$D \times 1.01) - 1000$$

After 2nd month

$$A_2 = (A_1 \times 1.01) - 1000$$

$$= [(\$D \times 1.01) - 1000] \times 1.01 - 1000$$

$$= \$D \times (1.01)^2 - 1000 \times 1.01 - 1000$$

$$= \$[D(1.01)^2 - 1000(1.01 + 1)]$$

$$= \$[D(1.01)^2 - 1000(1 + 1.01)] \quad (2)$$

$$D(1.01)^{120} - 1000[1.01^{119} + 1.01^{118} + 1.01^{117} + \dots + 1.01 + 1] \geq 0$$

this part is a series

$$a = 1, r = 1.01 \quad n = 120$$

$$S_n = \frac{1(1.01^{120} - 1)}{1.01 - 1}$$

$$D(1.01)^{120} - 1000 \times \frac{1(1.01^{120} - 1)}{1.01 - 1} > 0$$

$$D(1.01)^{120} > 1000 \times \frac{1(1.01^{120} - 1)}{1.01 - 1}$$

$$D > \frac{1000 \times (1.01^{120} - 1)}{1.01 - 1}$$

$$(1.01)^{120}$$

$$D > \$69,700 \text{ (nearest \$100)}$$

Extension 1 Year 11 Yearly 2009. Question 3.

a) $(-3, 4)$ $(5, -2)$ $1:3$
 x_1, y_1 x_2, y_2 $m:n$

Co-ords of point dividing

$$= \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{5 + 3(-3)}{4}, \frac{-2 + 3(4)}{4} \right)$$

$$= (-1, 2.5)$$

b)

$$y = 3x - 2 \quad m_1 = 3$$

$$x + 2y - 3 \quad m_2 = -1/2$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{3 - (-1/2)}{1 + (3)(-1/2)} \right|$$

$$\tan \theta = |-7| = 7$$

$$\theta = 81.87^\circ$$

$$\theta = 82^\circ \text{ (nearest degree)}$$

c)

$$f(-2) = 6(-2)^3 + 7(-2)^2 - 9(-2) + 2$$

$$= 0$$

$\therefore (x+2)$ is a factor

ii) Divide to find remaining factors.

$$\begin{array}{r} 6x^2 - 5x + 1 \\ x+2 \overline{) 6x^3 + 7x^2 - 9x + 2} \\ \underline{6x^3 + 12x^2} \\ -5x^2 - 9x \\ \underline{-5x^2 + 10} \\ x + 2 \end{array}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{5 \pm \sqrt{25 - 4(6)(1)}}{12}$$

$$= 1/3, 1/2$$

- but we need $6x^2$ so.

Factors are

$$6x^3 + 7x^2 - 9x + 2 = (x+2)(2x-1)(3x-1)$$

d)

$$\tan \theta = -1$$

$$\tan \theta = \tan(-\pi/4)$$

$$\therefore \theta = -\frac{\pi}{4} + n\pi \quad n \in \mathbb{Z}$$

e) i) $\cos \theta - \sqrt{3} \sin \theta = R \cos(\theta + \alpha)$

$$R = \sqrt{1^2 + \sqrt{3}^2} \quad \tan \alpha = \frac{\sqrt{3}}{1}$$

$$= 2$$

$$R \geq 0 \quad 0 \leq \alpha \leq \frac{\pi}{2} \quad \alpha = \pi/3$$

$$\therefore \cos \theta - \sqrt{3} \sin \theta = 2 \cos(\theta + \pi/3)$$

$$\begin{aligned} \text{ii) } \cos \theta - \sqrt{3} \sin \theta &= 1. \\ 2 \cos(\theta + \pi/3) &= 1. \\ \cos(\theta + \pi/3) &= 1/2 \end{aligned}$$

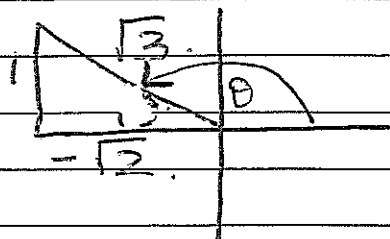
$$\text{as } 0 \leq \theta \leq 2\pi.$$

$$\pi/3 \leq \theta + \pi/3 \leq 2\pi + \pi/3.$$

$$\theta + \pi/3 = \pi/3, \frac{5\pi}{3}, \frac{7\pi}{3}.$$

$$\theta = 0, \frac{4\pi}{3}, 2\pi \quad \times$$

$$\text{f) } \sin \theta = \frac{1}{\sqrt{3}}, \frac{\pi}{2} < \theta < \pi.$$



$$\text{i) } \tan \theta = -\frac{1}{\sqrt{2}}.$$

$$\begin{aligned} \text{ii) } \cos 2\theta &= 1 - 2\sin^2 \theta. \\ &= 1 - 2\left(\frac{1}{\sqrt{3}}\right)^2. \\ &= +1/3 \quad \times. \end{aligned}$$

QUESTION FOUR

a) $\frac{x+4}{x-2} > 3 \quad x \neq 2$

$$\frac{x+4}{x-2} \cdot (x-2)^2 > 3(x-2)^2$$

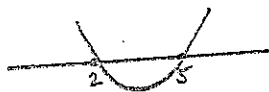
$$(x+4)(x-2) > 3(x-2)^2$$

$$x^2 + 2x - 8 > 3x^2 - 12x + 12$$

$$2x^2 - 14x + 20 \leq 0$$

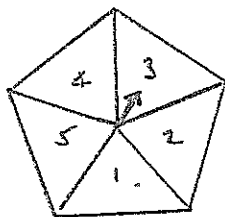
$$x^2 - 7x + 10 \leq 0$$

$$(x-5)(x-2) \leq 0$$



$$2 \leq x \leq 5$$

b)



Events are equiprobable

~~P(odd) which is only on~~

$$(1,1) \quad (1,2) \quad (1,3) \quad (1,4) \quad (1,5)$$

$$(2,1) \quad (2,2) \quad (2,3) \quad (2,4) \quad (2,5)$$

$$(3,1) \quad (3,2) \quad (3,3) \quad (3,4) \quad (3,5)$$

$$(4,1) \quad (4,2) \quad (4,3) \quad (4,4) \quad (4,5)$$

$$(5,1) \quad (5,2) \quad (5,3) \quad (5,4) \quad (5,5)$$

(i) $P(\text{odd sum}) = \frac{12}{25}$, $P(\text{at least 7}) = \frac{10}{25} = \frac{2}{5}$

ii) $P(\text{No 5 with 1 apen}) = \frac{4}{5}$

$$P(\text{No 5 with 2 apens}) = \left(\frac{4}{5}\right)^2$$

$$P(\text{No 5 with } n \text{ apens}) = \left(\frac{4}{5}\right)^n$$

$$P(\text{at least 1 FIVE}) = 1 - P(\text{no FIVES})$$

$$x = a(p+q)$$

So $1 - \left(\frac{4}{5}\right)^2 \approx 99.9\%$

$$1 - \left(\frac{4}{5}\right)^n \approx .999$$

$$-(.8)^n \approx .001$$

$$(.8)^n \leq .001$$

$$n \ln(.8) \leq \ln .001$$

$$n \geq \frac{\ln .001}{\ln .8}$$

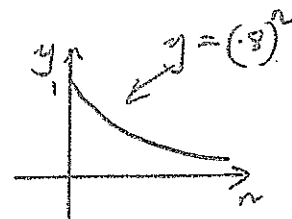
$$n \geq 30.9$$

$$n = 31$$

CHECK $n = 30 \Rightarrow .8^{30} = .00123 \dots$

$n = 31 \Rightarrow .8^{31} = .00099 \dots$

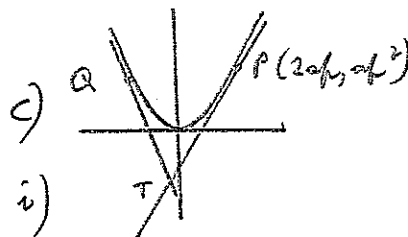
$$n = 31$$



NOTE

$\ln$ means $\log_e$

$\log_{10}$ could also be used.



$$x^2 = 4ap$$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

$$= \frac{2ap}{2a}$$

$$= p \text{ (when } x = 2ap)$$

OR $x = 2ap$
 $y = ap^2$

$$\frac{dy}{dx} = \frac{dy}{dp} \times \frac{dp}{dx}$$

$$= 2ap \times \frac{1}{2a}$$

$$= p$$

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2 \quad \text{--- (1)}$$

Tangent at Q is

$$y = qx - aq^2 \quad \text{--- (2)}$$

iii) Subtracting (1) - (2)

$$0 = x(p - q) - a(p^2 - q^2)$$

$$0 = x(p - q) - a(p - q)(p + q)$$

$$0 = x - a(p + q) \quad p \neq q$$

THEN

$$\begin{aligned} y &= fa - af^2 \\ &= fa(f+q) - af^2 \\ &= af^2 + afq - af^2 \\ &= afq \end{aligned}$$

So $x = a(f+q)$

$y = afq$

iii) NOW

$$\begin{aligned} SP &= \sqrt{(2af-0)^2 + (af^2-a)^2} \\ &= \sqrt{4a^2f^2 + a^2(f^4 - 2f^2 + 1)} \\ &= \sqrt{a^2f^4 + 2a^2f^2 + a^2} \\ &= \sqrt{(af^2+a)^2} \\ &= af^2 + a \end{aligned}$$

iv) $SQ = a(q^2+1)$

AND

$$SP + SQ = a(f^2+1) + a(q^2+1)$$

$$a(f^2+1) + a(q^2+1) = 4a$$

$$\Rightarrow f^2 + q^2 = 2$$

NOW

$$\frac{x}{a} = f+q \quad \frac{y}{a} = fq$$

$$\begin{aligned} \frac{x^2}{a^2} &= f^2 + 2fq + q^2 \\ &= (f^2 + q^2) + 2fq \end{aligned}$$

$$\frac{x^2}{a^2} = 2 + \frac{2y}{a}$$

$$\frac{x^2}{a^2} = 2 + \frac{2y}{a}$$

$$x^2 = 2ay + 2a^2$$

$$x^2 = 2a(y+a)$$

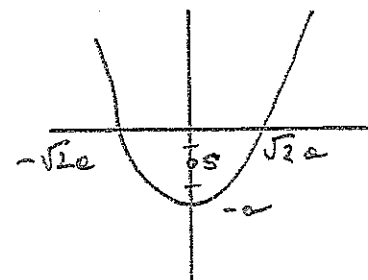
$$x^2 = 4 \cdot \frac{a}{2} (y+a)$$

$$X^2 = 4A \quad Y$$

Parabola

Vertex $(0, -a)$

Focal length $\frac{a}{2}$



Focus is point S